

Sets

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Creating a Set

Sets can be created empty or with initial data

When creating an empty set, you must use the `set()` function, not `{}` !

```
1 empty: set[int] = set()
2 has_initial_data: set[str] = {"a", "c", "b", "z"}
3
4 print(f"The empty set: {empty}")
5 print(f"The set with data: {has_initial_data}")
```

The empty set: set()

The set with data: {'c', 'a', 'z', 'b'}

Essential Set Methods

The following methods perform all of the basic operations we need when working with sets such as adding and removing data

Method	Description
<code>my_set.add(v)</code>	Adds the given value <code>v</code> to <code>my_set</code>
<code>my_set.discard(v)</code>	Removes the given value <code>v</code> from <code>my_set</code>
<code>my_set.remove(v)</code>	Removes the given value <code>v</code> from <code>my_set</code> , causes an error if <code>v</code> is not in <code>my_set</code>
<code>my_set.pop()</code>	Removes a random value from <code>my_set</code> , returns the removed value

Set Operations

Sets also have their own unique behaviors called set operations!

This category of operations are all performed between two sets:

We will break down each of these operations by comparing their results using a Venn Diagram of the following two sets A and B:

Set Union

Union

The union of two sets is a new set which contains the unique elements from both sets!

```
1 A: set[int] = {1, 2, 4, 8, 16}
2 B: set[int] = {1, 2, 3, 5, 8}
3
4 print("A | B:", A.union(B))
5 print("B | A:", B.union(A))
```

A | B: {1, 2, 3, 4, 5, 8, 16}

B | A: {1, 2, 3, 4, 5, 8, 16}

Set Intersection

Intersection

The intersection of two sets is a new set which only contains the shared elements from both sets!

```
1 A: set[int] = {1, 2, 4, 8, 16}
2 B: set[int] = {1, 2, 3, 5, 8}
3
4 print("A & B:", A.intersection(B))
5 print("B & A:", B.intersection(A))
```

A & B: {8, 1, 2}

B & A: {8, 1, 2}

Set Difference

Set Symmetric Difference

Set Subset

Subset

A set is a subset of another if all of its elements are contained by the other set!

```
1 A: set[int] = {1, 2, 4, 8, 16}
2 B: set[int] = {1, 2, 3, 5, 8}
3 C: set[int] = {1, 2, 3}
4
5 print("C <= A:", C.issubset(A))
6 print("C <= B:", C.issubset(B))
```

```
C <= A: False
C <= B: True
```

Set Superset

Superset

A set is a superset of another if it contains all of the elements of another set!

```
1 A: set[int] = {1, 2, 4, 8, 16}
2 B: set[int] = {1, 2, 3, 5, 8}
3 C: set[int] = {1, 2, 3}
4
5 print("A >= C:", A.issuperset(C))
6 print("B >= C:", B.issuperset(C))
```

```
A >= C: False
B >= C: True
```